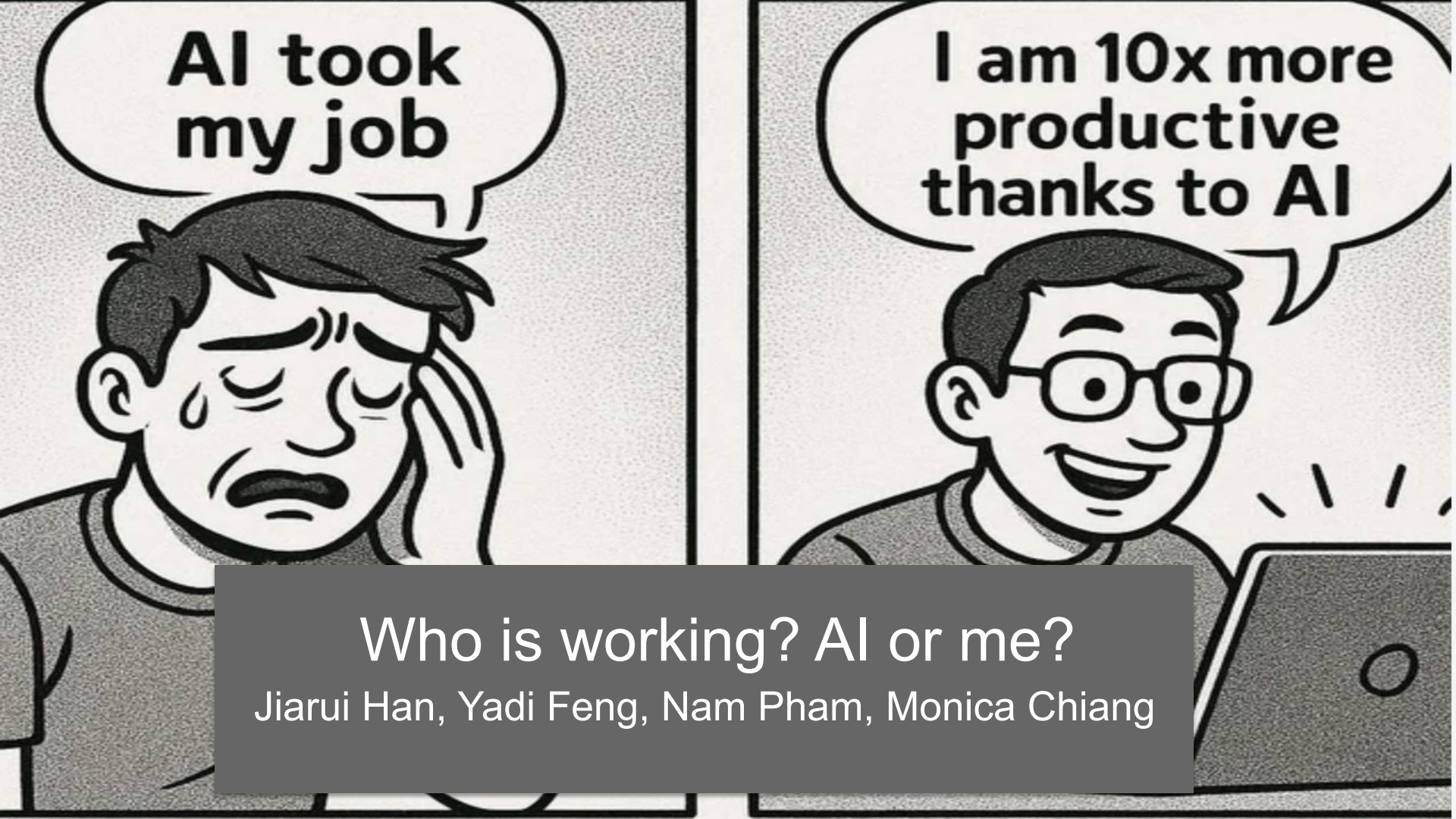




**AI took
my job**

**I am 10x more
productive
thanks to AI**

Who is working? AI or me?

Jiarui Han, Yadi Feng, Nam Pham, Monica Chiang



D6 - Presentation



D2 - Presentation

Topic Explanation

Why we chose this topic? How generative AI is reshaping workplace tasks

The rapid development of generative AI is reshaping how work is performed across industries. Modern AI systems can generate text, write and debug code, summarize documents, support decision-making, and perform analytical reasoning, increasingly embedded in everyday workflows.



Topic Explanation

Why we chose this topic? How generative AI is reshaping workplace tasks

The rapid development of generative AI is reshaping how work is performed across industries. Modern AI systems can generate text, write and debug code, summarize documents, support decision-making, and perform analytical reasoning, increasingly embedded in everyday workflows.

Does AI substitute for or complement human labor?



Topic Explanation

Why we chose this topic? How generative AI is reshaping workplace tasks

The rapid development of generative AI is reshaping how work is performed across industries. Modern AI systems can generate text, write and debug code, summarize documents, support decision-making, and perform analytical reasoning, increasingly embedded in everyday workflows.

Automation

- Autor (2015): Technology replaces specific tasks rather than entire occupations, leading to task reallocation rather than full job displacement.
- Acemoglu & Restrepo (2018, 2020): Formalize the distinction between automation (task substitution) and new task creation, labor market impact depends on this balance.

Substitution

- Brynjolfsson, Mitchell, & Rock (2018): ML systems substitute for some tasks while complementing others, depending on whether tasks are routine, data-intensive, or judgment-based.
- Eloundou et al. (2024): LLMs may affect a wide range of cognitive tasks, but exposure $\neq$ full automation.
- Felten, Raj, & Seamans (2021): Develop AI exposure measures; find substantial heterogeneity in how AI affects different types of work.

Topic Explanation

Why we chose this topic? How generative AI is reshaping workplace tasks

Empirical Evidence on Generative AI

Real-world studies suggest AI often enhances productivity rather than fully replacing workers

- Brynjolfsson, Li, & Raymond (2025): AI assistance in customer support improves worker productivity, particularly for less experienced employees — a strong augmentation effect.
- Noy & Zhang (2023): Generative AI significantly increases writing productivity and quality, complementarity between humans and AI in cognitive tasks.

Key challenge: Most studies focus on exposure or theoretical substitution, but less is known about how AI is actually used in practice (independent execution vs. collaborative support).

Topic Explanation

Why we chose this topic? How generative AI is reshaping workplace tasks

Two Fundamental modes of human-AI interaction



Automation

Where AI performs a task
with minimal human
involvement



Augmentation

Where AI supports, enhances, or
collaborates with human
workers

By focusing on observed AI usage patterns across occupations and tasks, this dashboard provides an empirical perspective on how AI is currently reshaping work in practice, rather than how it is expected to do so in theory.

Dataset Columns: Detailed Breakdown

All fields in the Anthropic Economic Index dataset organized by category

Task Variables

task_description

Full text description of the workplace task

task_type

Classification of the task category

incumbents_responding

Number of workers who confirmed this task

Task Metadata					
task_id	task_name	task_type	incumbents_responding	date	domain_source
13077	create music	Core	82	06/2007	Incumbent
5682	to student	Core	95	07/2011	Incumbent
5700	to student	supplemental	58	07/2011	Incumbent
5726	to student	supplemental	107	07/2011	Incumbent
5751	to student	supplemental	72	07/2011	Incumbent
5774	to student	supplemental	70	07/2011	Incumbent
5793	to student	supplemental	90	07/2011	Incumbent
5818	to student	supplemental	118	07/2011	Incumbent
5843	to student	supplemental	111	07/2011	Incumbent
5866	to student	supplemental	63	07/2011	Incumbent
5888	to student	Core	102	07/2011	Incumbent
5914	to student	supplemental	92	07/2011	Incumbent
5934	to student	Core	48	07/2011	Incumbent
5960	to student	Core	58	07/2011	Incumbent
5980	to student	Core	52	07/2011	Incumbent
6005	to student	supplemental	103	07/2011	Incumbent
6029	to student	supplemental	56	07/2011	Incumbent
6049	to student	supplemental	62	07/2011	Incumbent
6072	to student	supplemental	84	07/2011	Incumbent
6095	to student	supplemental	76	07/2011	Incumbent
6117	to student	Core	59	07/2011	Incumbent
6144	to student	supplemental	67	07/2011	Incumbent
6169	to student	supplemental	93	07/2011	Incumbent
6191	to student	supplemental	99	07/2011	Incumbent
6214	to student	Core	103	07/2011	Incumbent

overall_task_ai_usage

Aggregate AI usage at the task level

occupation_task_ai_usage

AI usage for a specific occupation–task pair

directive

AI directly performs the task (automation)

feedback_loop

AI reviews and provides feedback (automation)

task iteration

AI helps refine and iterate work (augmentation)

validation

AI checks and validates outputs (augmentation)

learning

AI teaches or explains concepts (augmentation)

filtered

Non-work or irrelevant AI activity

[illegible]

Dataset Columns: Detailed Breakdown

All fields in the Anthropic Economic Index dataset organized by category

Occupation Variables

job_name

Standard occupational title

soc_code

Standard Occupational Classification code

job_family

Broad occupational category grouping

job_zone

O*NET Job Zone (1 = little preparation to 5 = extensive)

median_salary

Median annual wage for the occupation

is_bright

Whether O*NET designates Bright Outlook status

is_green

Whether classified as environmentally sustainable

job_forecast

Projected job openings from BLS

				Occupational Metadata				
soc_code	soc_major_group	job_name	job_family	is_bright	is_green	job_zone	median_salary	job_forecast
27-2041.04	27	osers and Entertainment, Spo		FALSE	FALSE	3	49630	7800
25-1011.0C	25	achers, Pcon, Training, and		TRUE	FALSE	5	83960	11200
25-1021.0C	25	e Teacherson, Training, and		FALSE	FALSE	5	82220	3600
25-1022.0C	25	nce Teacheron, Training, and		FALSE	FALSE	5	73230	5300
25-1031.0C	25	Teachers, on, Training, and		TRUE	FALSE	5	86980	800
25-1032.0C	25	eachers, on, Training, and		TRUE	FALSE	5	101720	4700
25-1041.0C	25	ces Teachon, Training, and		FALSE	FALSE	5	84640	1200
25-1042.0C	25	ce Teacheron, Training, and		TRUE	FALSE	5	82550	6400
25-1043.0C	25	n Science on, Training, and		FALSE	FALSE	5	86900	200
25-1051.0C	25	d Space Scon, Training, and		FALSE	FALSE	5	90860	1200
25-1052.0C	25	achers, Pon, Training, and		FALSE	FALSE	5	79550	2400
25-1053.0C	25	nce Teachon, Training, and		FALSE	FALSE	5	79910	700
25-1054.0C	25	chers, Poon, Training, and		TRUE	FALSE	5	90800	1600
25-1061.0C	25	eology Teon, Training, and		TRUE	FALSE	5	83940	700
25-1062.0C	25	al Studieson, Training, and		TRUE	FALSE	5	74440	1200
25-1063.0C	25	achers, Pon, Training, and		TRUE	FALSE	5	101480	1500
25-1064.0C	25	achers, Pon, Training, and		FALSE	FALSE	5	80350	400
25-1065.0C	25	e Teacheron, Training, and		TRUE	FALSE	5	83370	1900
25-1066.0C	25	eachers, Fon, Training, and		TRUE	FALSE	5	76710	4800
25-1067.0C	25	achers, Pon, Training, and		FALSE	FALSE	5	74140	1600
25-1071.0C	25	es Teacheron, Training, and		TRUE	FALSE	5	97370	29400
25-1072.0C	25	and Teaclon, Training, and		TRUE	FALSE	5	73490	7700
25-1081.0C	25	achers, Pon, Training, and		TRUE	FALSE	5	64780	7400
25-1082.0C	25	Teacherson, Training, and		FALSE	FALSE	5	71560	500
25-1111.0C	25	Enforcemeron, Training, and		TRUE	FALSE	5	61900	1700
25-1112.0C	25	ers, Post on, Training, and		TRUE	FALSE	5	111140	2200
25-1113.0C	25	eachers, on, Training, and		TRUE	FALSE	5	68300	1500
25-1121.0C	25	asic Teachon, Training, and		TRUE	FALSE	5	69960	11100
25-1122.0C	25	Teachers, on, Training, and		FALSE	FALSE	5	68910	3200
25-1123.0C	25	terature on, Training, and		FALSE	FALSE	5	66590	7300
25-1124.0C	25	terature on, Training, and		TRUE	FALSE	5	67640	3000
25-1125.0C	25	chers, Poon, Training, and		FALSE	FALSE	5	74590	2400
25-1126.0C	25	gion Teaclon, Training, and		TRUE	FALSE	5	71890	3100
25-1192.0C	25	Teachers, on, Training, and		FALSE	FALSE	5	71380	200
25-1193.0C	25	Studies Ion, Training, and		FALSE	FALSE	5	62620	1800
25-9021.0C	25	ie Managemon, Training, and		FALSE	TRUE	5	49840	1000
41-9021.0C	41	Estate Br Sales and Related		TRUE	FALSE	4	58210	10100
41-9022.0C	41	tate Sales Sales and Related		TRUE	FALSE	3	48690	38900
11-9141.0C	11	nd Commun Management		TRUE	FALSE	4	58340	31600
25-9031.0C	25	primary on, Training, and		FALSE	FALSE	5	64450	18600

Initial questions



to explore

Question 1

Where is generative AI being used most?

Which occupations and workplace tasks have the highest levels of generative AI exposure?

Highlights

- Identifies where generative AI is currently being used most frequently
- Examines differences in AI exposure across occupational families, occupations, and individual workplace tasks
- Reveals which specific activities appear most often in AI-related workplace conversations

Question 2

Is AI a worker or a copilot?

Is AI primarily replacing human task execution or supporting human workers across different occupations?

■ Automation: AI as an Independent Worker

- AI directly performs or completes the task
- Captured by: directive + feedback_loop
- AI acts largely on its own with minimal human involvement

■ Augmentation: AI as a Human Copilot

- AI assists, validates, teaches, or collaborates with the worker
- Captured by: task_iteration + validation + learning
- AI supports workers through collaboration and iterative improvement

Question 3

Does the automation–augmentation balance vary across occupational groups?

Does the balance between automation and augmentation change across job-preparation levels (Job Zones) or salary groups?

Section A: Comparison across Job Zones

Job Zones represent the level of education, experience, and training required for an occupation (O*NET scale 1–5):

- Job Zone 1: Little or no preparation
- Job Zone 2: Some preparation
- Job Zone 3: Medium preparation
- Job Zone 4: High preparation
- Job Zone 5: Extensive preparation

Section B: Comparison across Salary Groups

Occupations divided into salary quartiles for balanced group comparison:

- Lower salary (Q1)
- Lower-middle salary (Q2)
- Upper-middle salary (Q3)
- Higher salary (Q4)



D4 - Presentation

Previous Presentation

Which occupations and workplace tasks have the highest levels of generative AI exposure?

Is AI primarily replacing human task execution or supporting human workers across different occupations?

Does the balance between automation and augmentation change across job-preparation levels (Job Zones) or salary groups?

Data Cleaning

Statistics on Missing Values and Data Types

column_name	interpreted_type	total_count	missing_count	missing_percentage	unique_count	example_values
task_id	float	4087	1	0.02	4086	['13077.0', '5682.0', '5700.0']
task_name	string/object	4087	1	0.02	3365	['Accept commissions to create music for special occasions.', 'Act as advisers to student']
task_type	string/object	4087	124	3.03	2	['Core', 'Supplemental']
incumbents_responding	float	4087	124	3.03	154	['82.0', '95.0', '58.0']
date	string/object	4087	1	0.02	17	['06/2007', '07/2011', '07/2014']
domain_source	string/object	4087	1	0.02	3	['Incumbent', 'Occupational Expert', 'Analyst']
task_percentage_all	float	4087	0	0	661	['0.002054156', '0.004827266', '0.001643325']
num_jobs_sharing_task	float	4087	1	0.02	18	['1.0', '34.0', '6.0']
task_percentage_job	float	4087	1	0.02	699	['0.002091418', '0.0001445538892', '0.001673134']
task_name_normalized	string/object	4087	1	0.02	3364	['accept commissions to create music for special occasions.', 'act as advisers to student']
feedback_loop	float	4087	1	0.02	469	['0.0', '0.010558069', '0.014285714']
directive	float	4087	1	0.02	1376	['0.0', '0.382978723', '0.391304348']
task_iteration	float	4087	1	0.02	1295	['0.0', '0.361702128', '0.434782609']
validation	float	4087	1	0.02	358	['0.0', '0.018602313', '0.032539683']
learning	float	4087	1	0.02	1292	['0.0', '0.060331825', '0.134920635']
filtered	float	4087	1	0.02	1543	['1.0', '0.255319149', '0.173913044']
soc_code	string/object	4087	1	0.02	749	['27-2041.04', '25-1011.00', '25-1021.00']
soc_major_group	float	4087	1	0.02	22	['27.0', '25.0', '41.0']
job_name	string/object	4087	1	0.02	749	['Music Composers and Arrangers', 'Business Teachers, Postsecondary', 'Computer Science']
job_family	string/object	4087	1	0.02	22	['Arts, Design, Entertainment, Sports, and Media', 'Education, Training, and Library', '']
is_bright	string/object	4087	1	0.02	2	['False', 'True']
is_green	string/object	4087	1	0.02	2	['False', 'True']
job_zone	float	4087	1	0.02	6	['3.0', '5.0', '4.0']
median_salary	float	4087	1	0.02	556	['49630.0', '83960.0', '82220.0']
job_forecast	float	4087	1	0.02	290	['7800.0', '11200.0', '3600.0']
chance_auto	float	4087	1	0.02	110	['-1.0', '0.8', '97.0']
wage_group	string/object	4087	2781	68.05	111	['Music Directors and Composers', 'Instructional Coordinators', 'Writers and Authors']

Number of rows: 4,087 (before cleaning) → 3,963 (after cleaning)

Data Cleaning

Variables Definition

Automation

Definition

feedback_loop

[Automation] How often AI finishes tasks with feedback/corrections. Ratio indicating feedback loop automation pattern (0-1)

directive

[Automation] How often AI just gets a task and does it right away. Ratio indicating directive automation pattern (0-1)

Augmentation

task_iteration

[Augmentation] How often AI and user work together and improve results. Ratio indicating task iteration augmentation pattern (0-1)

validation

[Augmentation] How often AI checks or reviews users' work. Ratio indicating validation augmentation pattern (0-1)

learning

[Augmentation] How often AI explains, teaches, or helps build user skills. Ratio indicating learning augmentation pattern (0-1)

Data Cleaning

Variables Construction

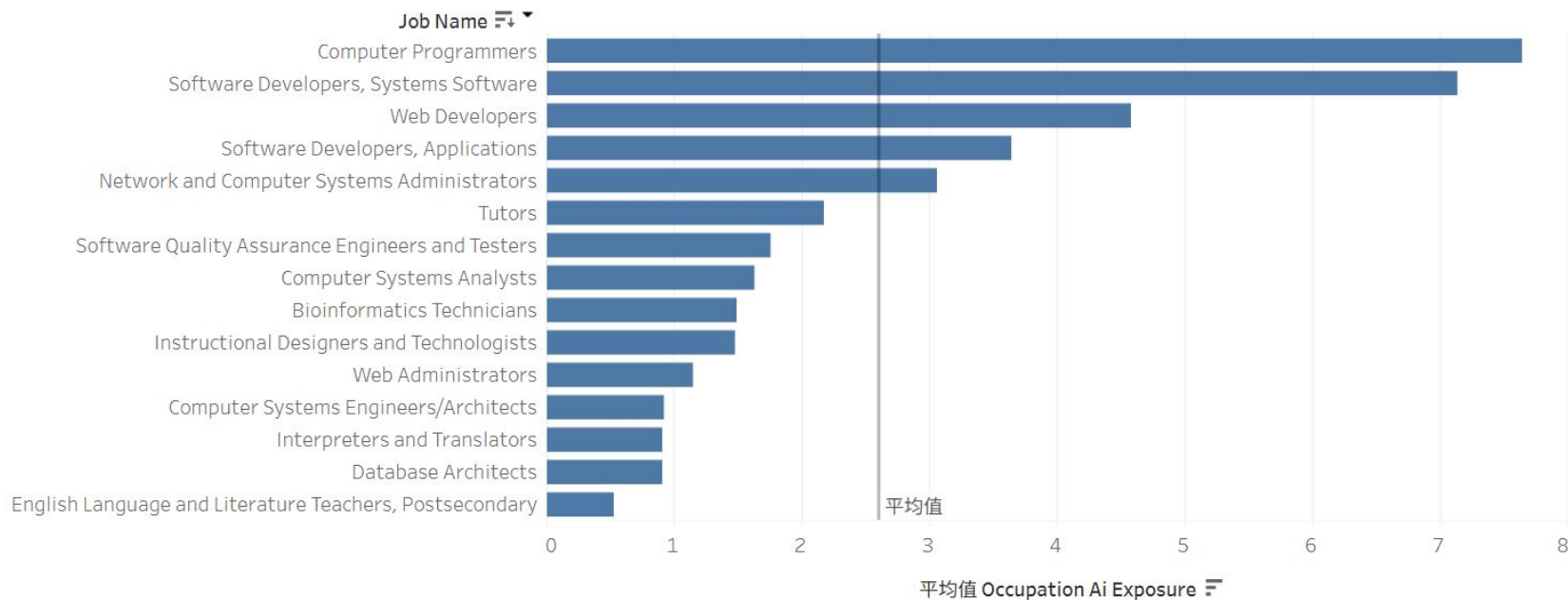
New Variables	Level	Definition
occupation_ai_exposure	Occupation	The sum of task_percentage_job for all tasks of the same occupation.
task_automation_raw	Task	directive + feedback_loop
task_augmentation_raw	Task	task_iteration + validation + learning
valid_activity_share	Task	1 - filtered (How much was filtered out as not related to jobs/work)
auto_augmentation_sum	Task	Used to check if the sum of two indicators is 1.
orientation_sum_check	Task	Is the sum of the indicators close to 1? True/False

Question1 explored: Where Is AI Being Used?

Which professions have the highest level of generative AI usage?

Which professions have the highest level of generative AI usage?

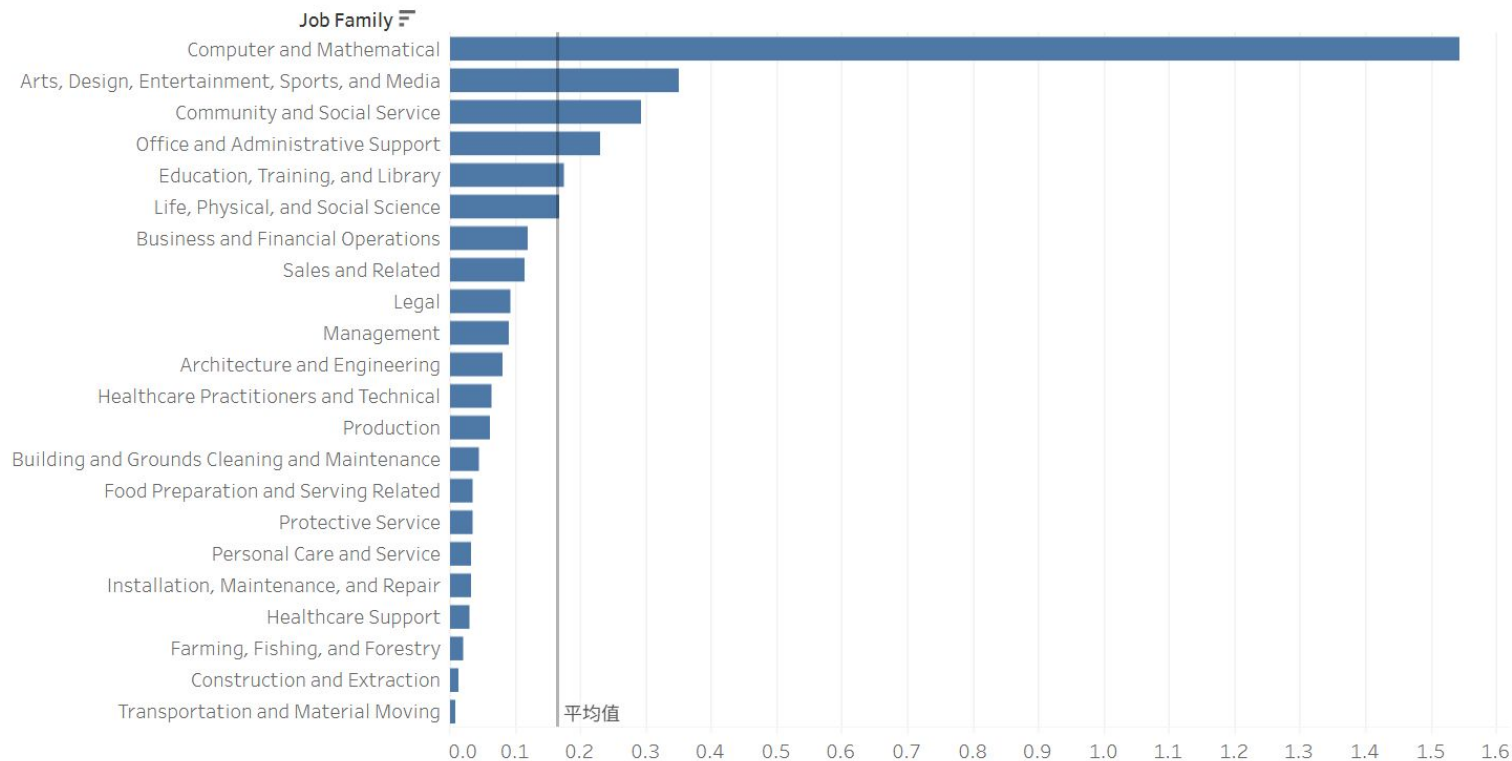
TOP 15 Jobs



Question1 explored: Where Is AI Being Used?

Which broad occupational families have the highest average AI exposure?

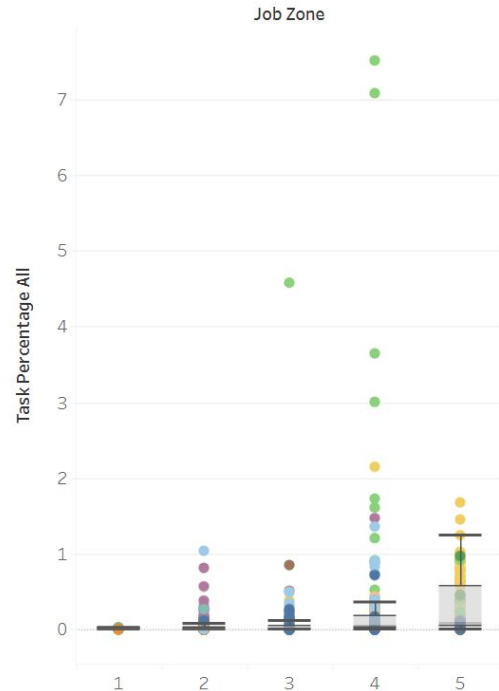
Which broad occupational families have the highest average AI exposure?



Question1 explored: Where Is AI Being Used?

Is AI exposure concentrated in occupations requiring higher levels of education, experience, and training?

Is AI exposure concentrated in occupations requiring higher levels of education, experience, and training?



Percentage (0–100) of all AI conversations associated with this task. Measures how frequently people use AI for this task overall.

$P(\text{task})$

Job Family

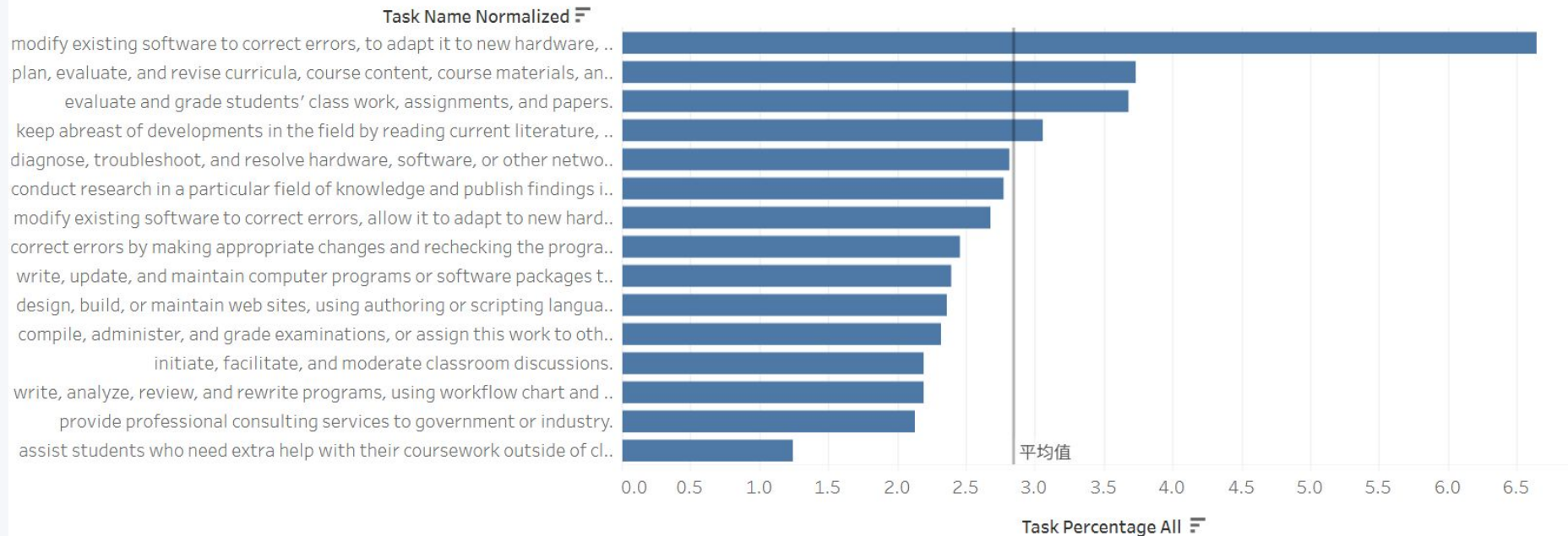
- Architecture and Eng...
- Arts, Design, Enterta...
- Building and Ground...
- Business and Financi...
- Community and Soci...
- Computer and Mathe...
- Construction and Ext...
- Education, Training, ...
- Farming, Fishing, and...
- Food Preparation an...
- Healthcare Practitio...
- Healthcare Support
- Installation, Mainten...
- Legal
- Life, Physical, and So...
- Management
- Office and Administr...
- Personal Care and Se...
- Production
- Protective Service
- Sales and Related
- Transportation and ...

Question1 explored: Where Is AI Being Used?

Which specific tasks appear most frequently in AI conversations?

Which specific tasks appear most frequently in AI conversations?

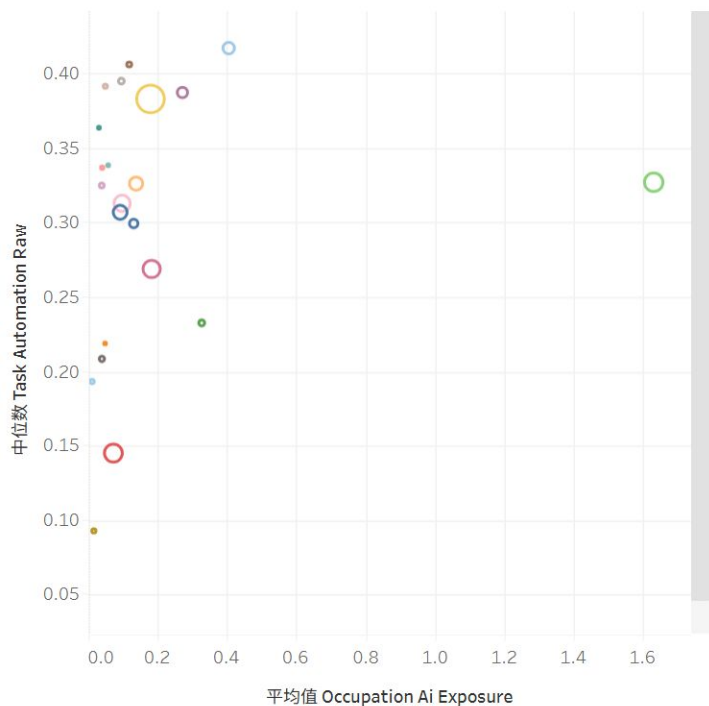
TOP 15 tasks



Question2 explored: Is AI a Worker or a Copilot?

For professions with high AI exposure, does the role involve AI directly performing tasks, or AI assisting humans?

For professions with high AI exposure, does the role involve AI directly performing tasks, or AI assisting humans?



Question2 explored: Is AI a Worker or a Copilot?

Looking at AI work activities overall, is AI more about automation or augmentation?

Visualization 6: Overall Automation versus Augmentation Share

分析问题： 从总体观察到的 AI 工作活动来看，AI 更多是 automation 还是 augmentation?

设置	内容
图表	100% stacked bar
分类	Automation、Augmentation
数值	Exposure-weighted overall share
推荐	使用单个横向 100% stacked bar
不推荐	复杂 pie chart 或多个 gauge

Question2 explored: Is AI a Worker or a Copilot?

Which occupational families are more prone to automation, and which are more prone to augmentation?

Visualization 7: Automation–Augmentation Balance by Occupational Family

分析问题： 哪些 occupational families 更倾向于 automation，哪些更倾向于 augmentation？

设置

内容

图表

100% stacked horizontal bar

Y 轴

Occupational family

两部分

Automation、Augmentation

排序

按 Automation share 从高到低

分析单位

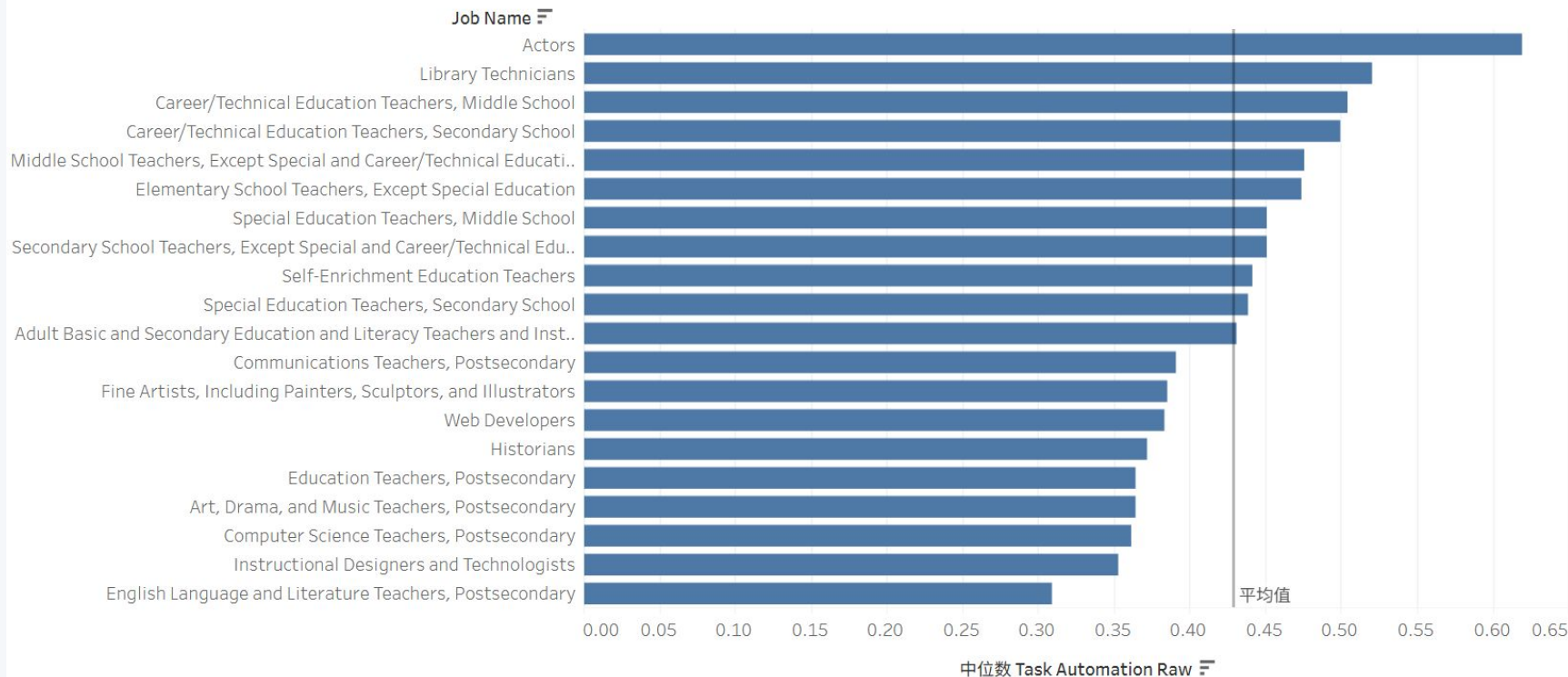
先计算 occupation-level，再聚合到 family

Question2 explored: Is AI a Worker or a Copilot?

Which professions are most inclined to have AI directly execute or complete tasks?

Which professions are most inclined to have AI directly execute or complete tasks?

TOP 20 Jobs

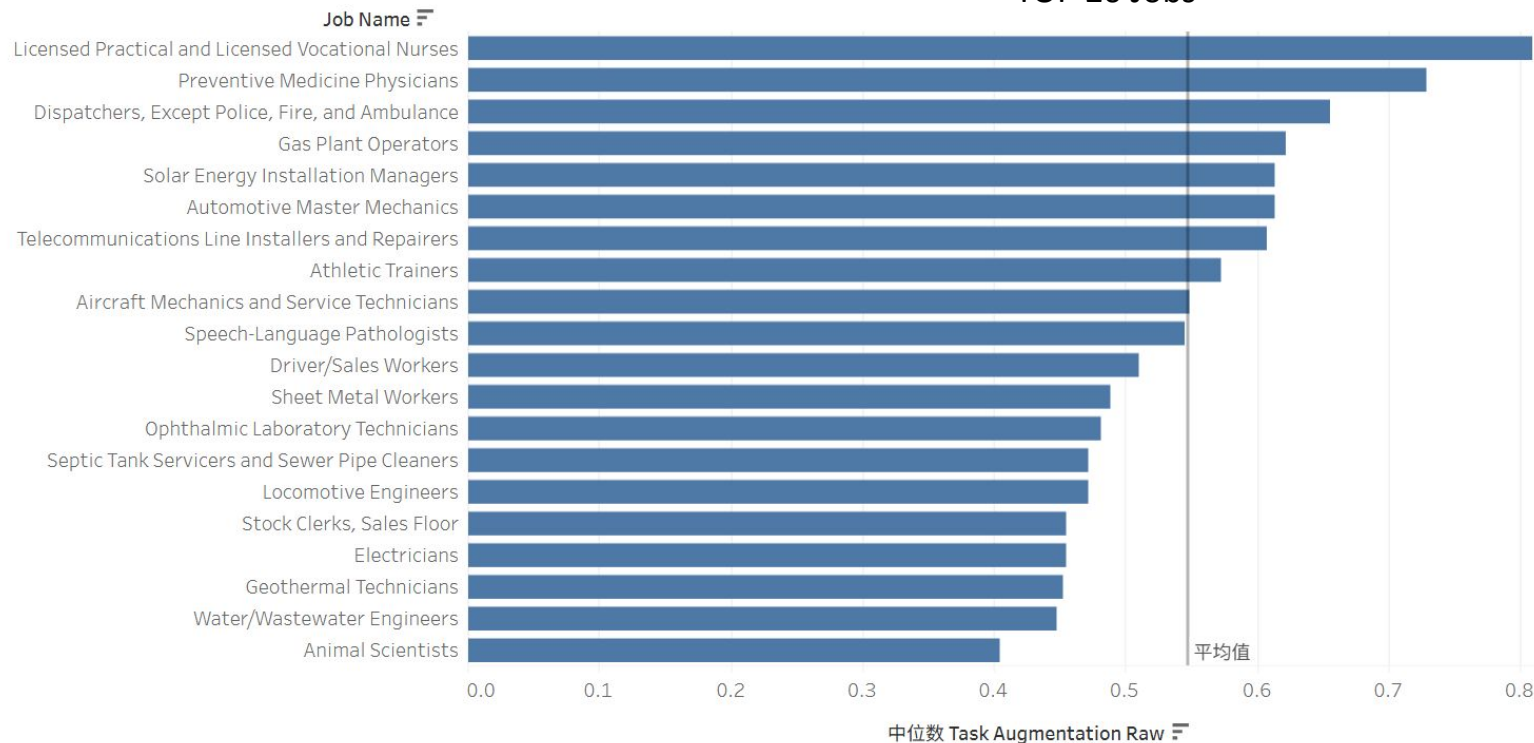


Question2 explored: Is AI a Worker or a Copilot?

Which professions are most inclined to use AI as a copilot, validator, or learning tool?

Which professions are most inclined to use AI as a copilot, validator, or learning tool?

TOP 20 Jobs



Question2 explored: Is AI a Worker or a Copilot?

What specific forms of interaction actually constitute automation and augmentation?

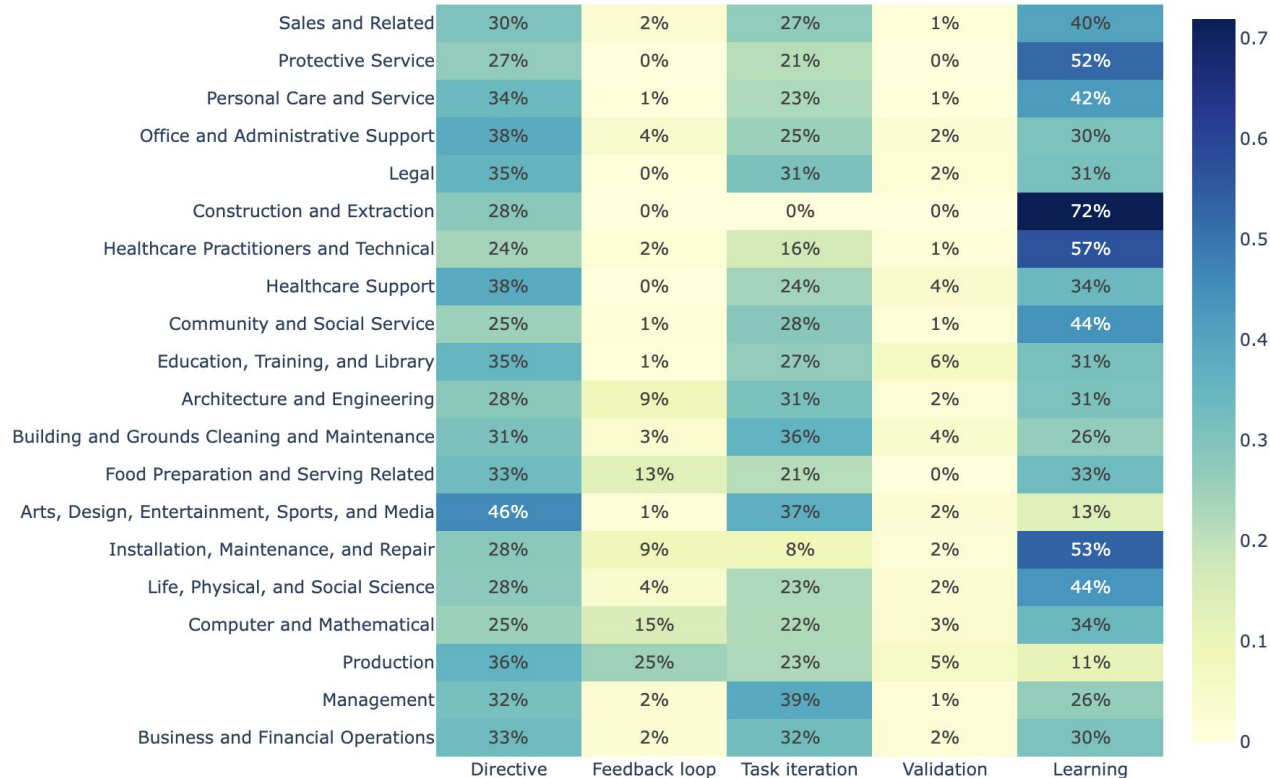
What specific forms of interaction actually constitute automation and augmentation?



Question2 explored: Is AI a Worker or a Copilot?

Which specific interaction modes drive automation and augmentation?

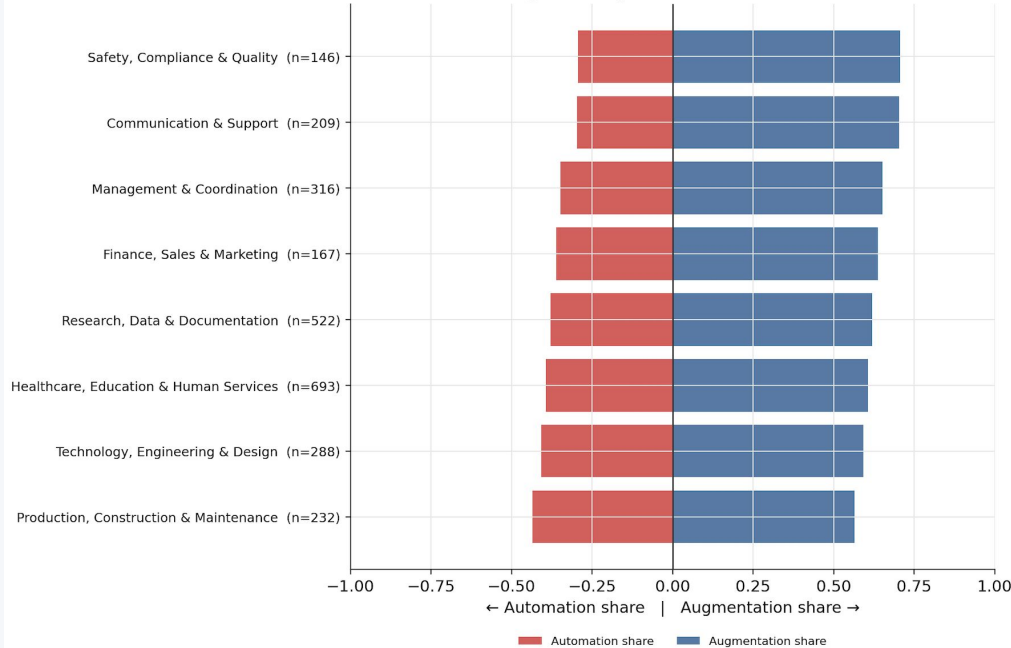
2.3 What kind of AI interaction drives exposure, by job family



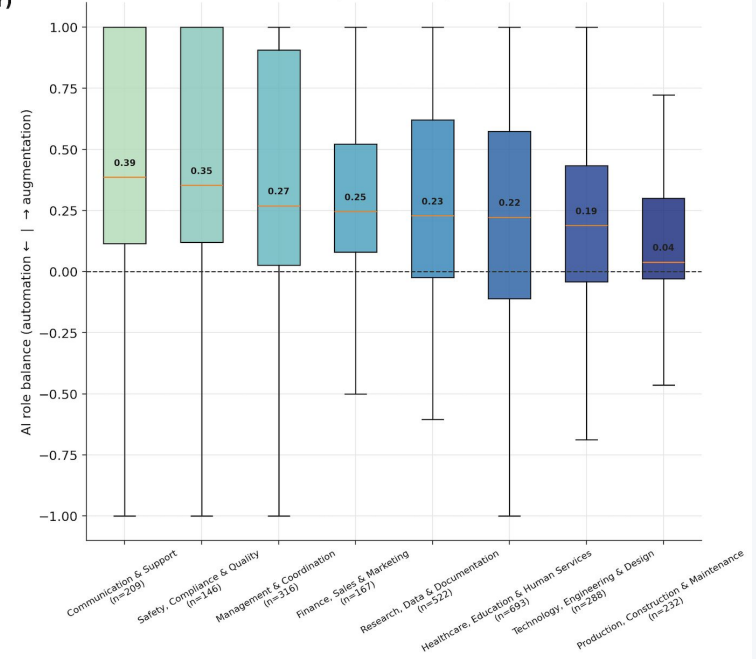
Question2 explored: Is AI a Worker or a Copilot?

Does the same occupation use AI differently across its individual tasks?

2.4b AI role balance across all task_clusters
(whole dataset, average share per cluster; n = valid tasks in that cluster)



AI role balance across all 8 task_clusters
(whole dataset)



Thank you
for
listening 😊